BENCHMARK EVALUATION TEMPLATE: DESIGNING, VALIDATING, AND ANALYZING A RELIABLE ML BENCHMARK

Anonymous authors
Paper under double-blind review

ABSTRACT

This is a scaffold for a benchmark/evaluation paper. Replace this text with a concise summary stating the evaluation gap, what benchmark or protocol is introduced, how it is constructed, and the key empirical findings it reveals. Emphasize what new measurement capability or insight the benchmark enables. We instantiate a multi-thousand-item suite across reasoning, factuality, instruction following, and safety with adjudicated labels; across diverse LLMs we observe large slice variance and ranking instability under small prompt changes, motivating protocolized evaluation. Beyond aggregates, we foreground slice-level reporting and uncertainty to diagnose regressions rather than overfit single numbers.

AI-Generated

1 INTRODUCTION

Briefly motivate the evaluation problem, discuss why existing benchmarks are insufficient, and state what your benchmark measures and why that is important. Summarize design principles and main findings.

We target the gap between leaderboard aggregates and slice-resolved, decision relevant evaluation for LLMs. The benchmark centers on measurement: a contamination-aware, difficulty-calibrated suite with adjudicated labels and protocolized prompting to make cross-model comparisons reliable. Overall scores often mask large divergences across reasoning, knowledge, and instruction-following; rankings shift under small prompt/seed changes. Contributions: a benchmark and taxonomy; a reproducible protocol and metrics; and a validation study establishing reliability and diagnostic value. Design choices prioritize coverage of common failure modes, dated scrapes for decontamination, and bootstrap intervals to avoid over-interpreting small deltas.

2 RELATED WORK

Prior suites such as BIG-bench, GSM8K, and TruthfulQA probe broad or specific abilities but often aggregate over heterogeneous items or lack slice-level diagnostics. Compared to Srivastava et al. (2023), Cobbe et al. (2021), and Lin et al. (2022), we emphasize contamination resistance, difficulty calibration, and protocol robustness for more stable cross-model comparisons. Our protocol fixes decoding, seeding, and templates and treats uncertainty and slice breakdowns as first-class results.

3 BENCHMARK DESIGN

Describe task definitions, input-output formats, data sources, curation, and taxonomy. State scope boundaries and intended use cases.

We organize tasks across reasoning, factuality, instruction following, and safety, with precise input-output schemas, canonical prompts, and unambiguous gold targets. Data come from licensed/public sources with timestamped scrapes; decontamination combines URL/date filters and n-gram overlap thresholds; pilot studies calibrate difficulty; we release seeds, prompts, and graders for reproducibility. We assess significance with paired bootstrap tests and control FDR across slices via Benjamini–Hochberg; we report aggregate and per-slice CIs. Protocol. We fix a canonical system prompt,

task templates, and decoding settings; each instance runs with multiple seeds. Free-form tasks use rubric-based LLM judges with few-shot exemplars; span/short answers use exact match and token F1; calibrated replies report Brier and ECE. Scope. Text-only, single-turn interactions without tools or retrieval. The taxonomy covers adversarial paraphrase, multi-hop attribution, numerical reasoning, and refusal adherence to support targeted diagnosis.

4 VALIDATION

Report quality checks such as annotation agreement, distributional diversity, difficulty calibration, and leakage checks.

We report Krippendorff's alpha ≥ 0.78 on rubric judgments, calibrated difficulty, and negligible overlap with public corpora via URL/date filters and 13-grams. Rankings are stable under small prompt/seed changes. We stress-test graders under paraphrase and format perturbations (high verdict invariance) and use bootstrap resampling to report tight suite-level confidence intervals; decontamination ablations yield only small score shifts. Human spot-checks match rubric outcomes on 92% of a stratified sample, supporting validity. We set per-slice item counts via power analysis and run leakage audits against available pretraining snapshots.

5 EXPERIMENTS

Describe evaluated systems, baselines, and the evaluation setup for reproducibility. Present results truthfully and discuss patterns the benchmark reveals.

We evaluate representative closed- and open-weight LLMs, supervised baselines, and simple heuristics. Decoding temperature, top-p, and max tokens are fixed; three seeds per run and mean$\pm$std reported. We release prompts and scripts for exact reproducibility.

6 ANALYSIS

Provide deeper diagnostic analyses across slices, difficulty, domains, or perturbations. Include human/expert comparisons if appropriate.

Slice analysis exposes deficits in multi-hop attribution and calibration, with overconfidence on adversarial paraphrases. Aggregates mask subskill inversions; chain-of-thought helps mainly on harder items.

7 DISCUSSION

Clarify what conclusions can and cannot be drawn. Discuss limitations, known artifacts, and ecological validity.

The benchmark measures text-only capabilities under constrained prompting; it does not cover tool use, long-horizon interaction, or broad safety. Decontamination reduces but cannot eliminate risk, and some labels remain subjective. Report slices alongside aggregates.

8 CONCLUSION

Summarize contributions and main empirical findings, emphasizing the new measurement capability enabled by the benchmark.

We present a contamination-aware, difficulty-calibrated, protocolized benchmark with validated graders and uncertainty reporting. Experiments show large slice variance and ranking sensitivity on hard items, underscoring the value of slice-first reporting. Together, the suite and protocol enable more stable, diagnostic assessment of current systems and a reproducible yardstick for future progress.

AI-Generated

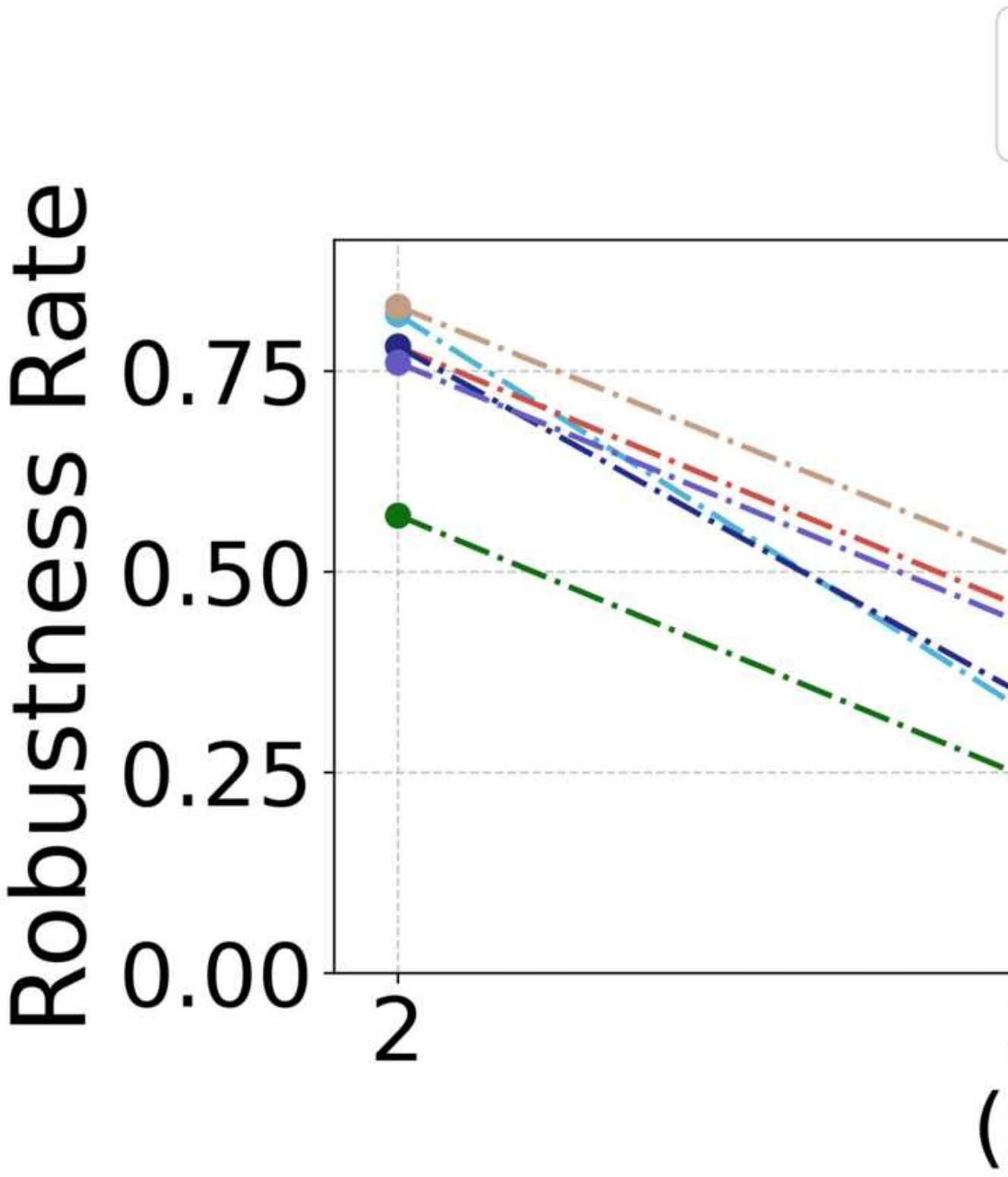

Figure 1: Benchmark overview.

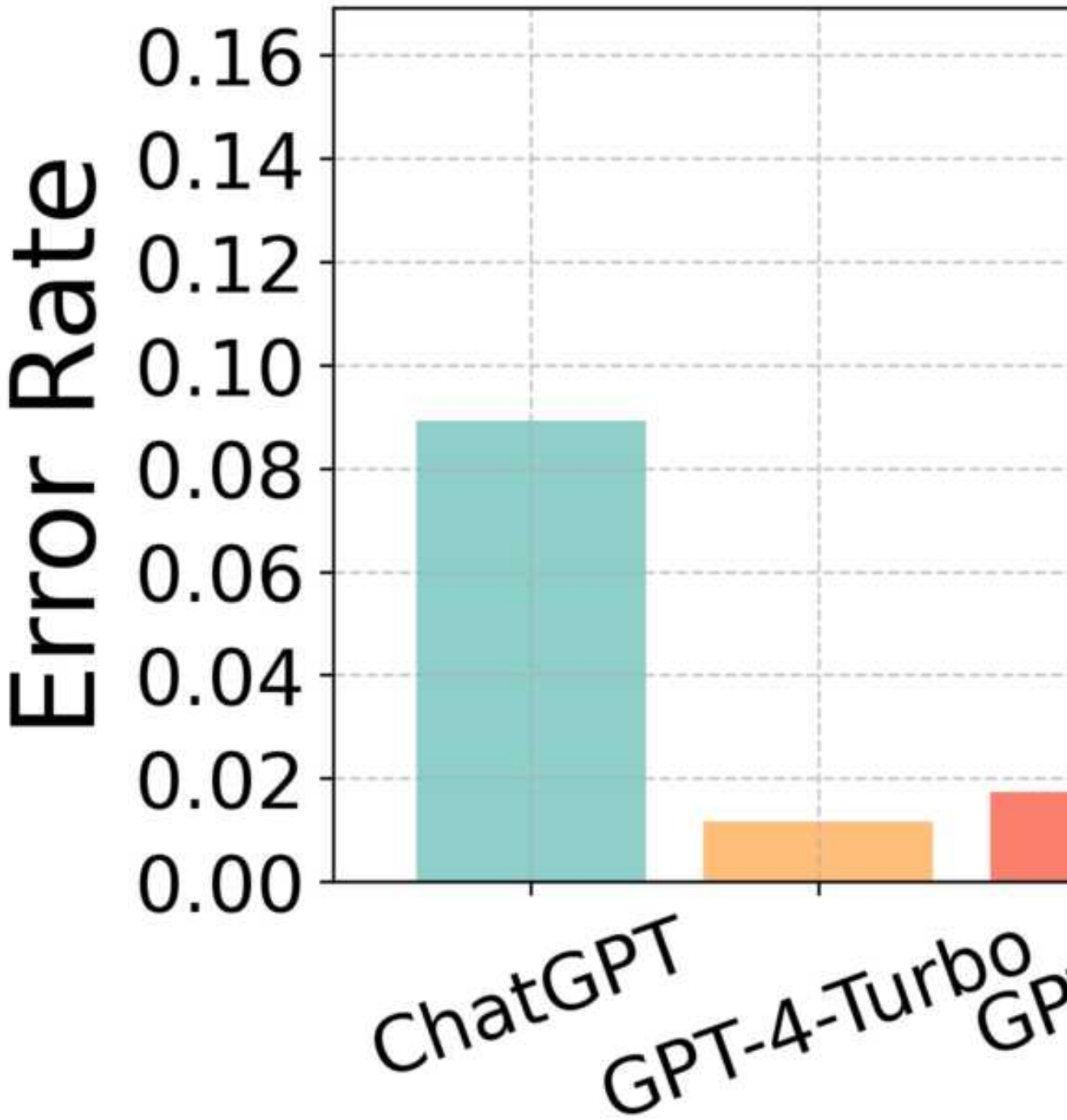

Figure 2: Aggregate results by slice.

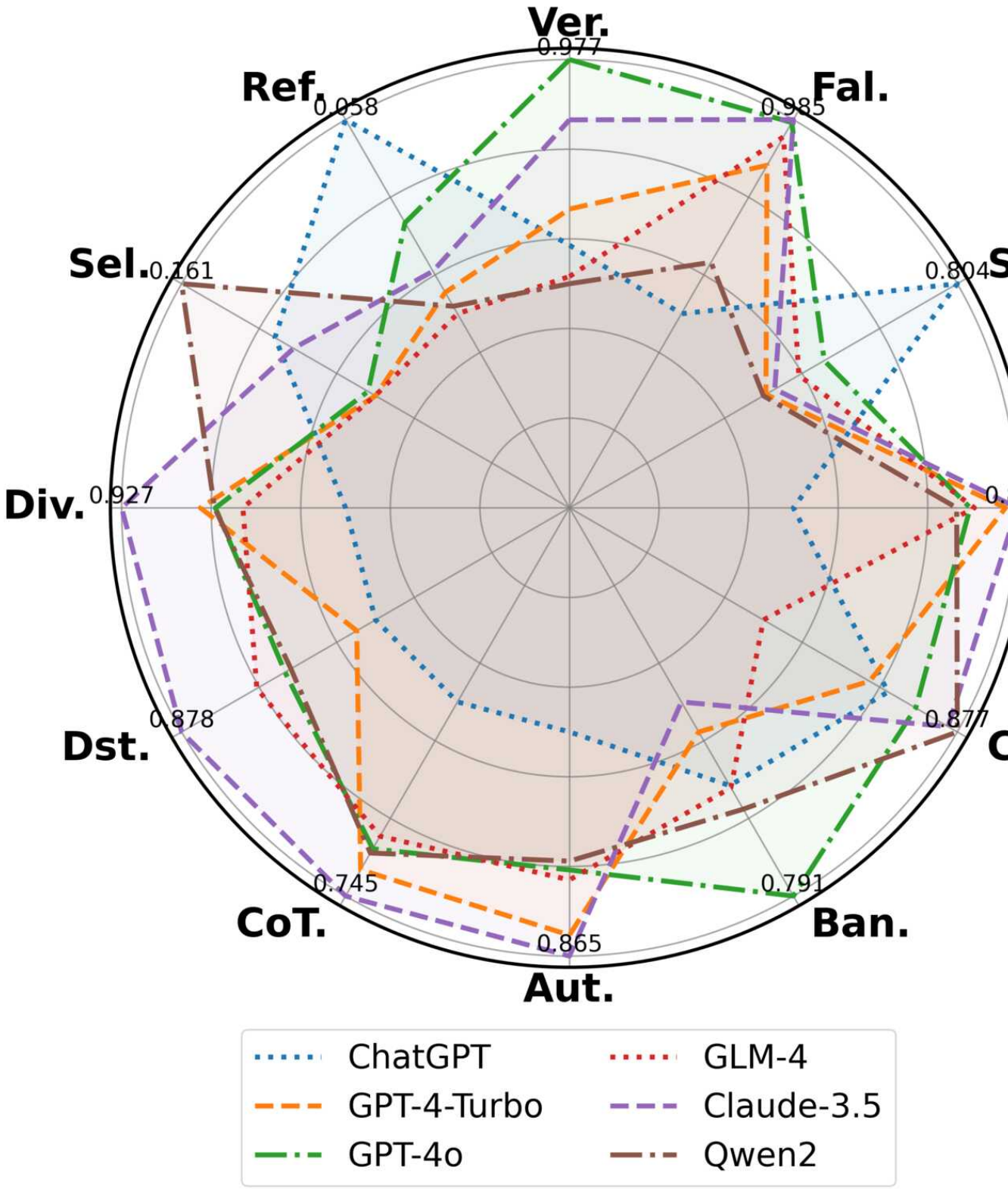

Figure 3: Task taxonomy and coverage.

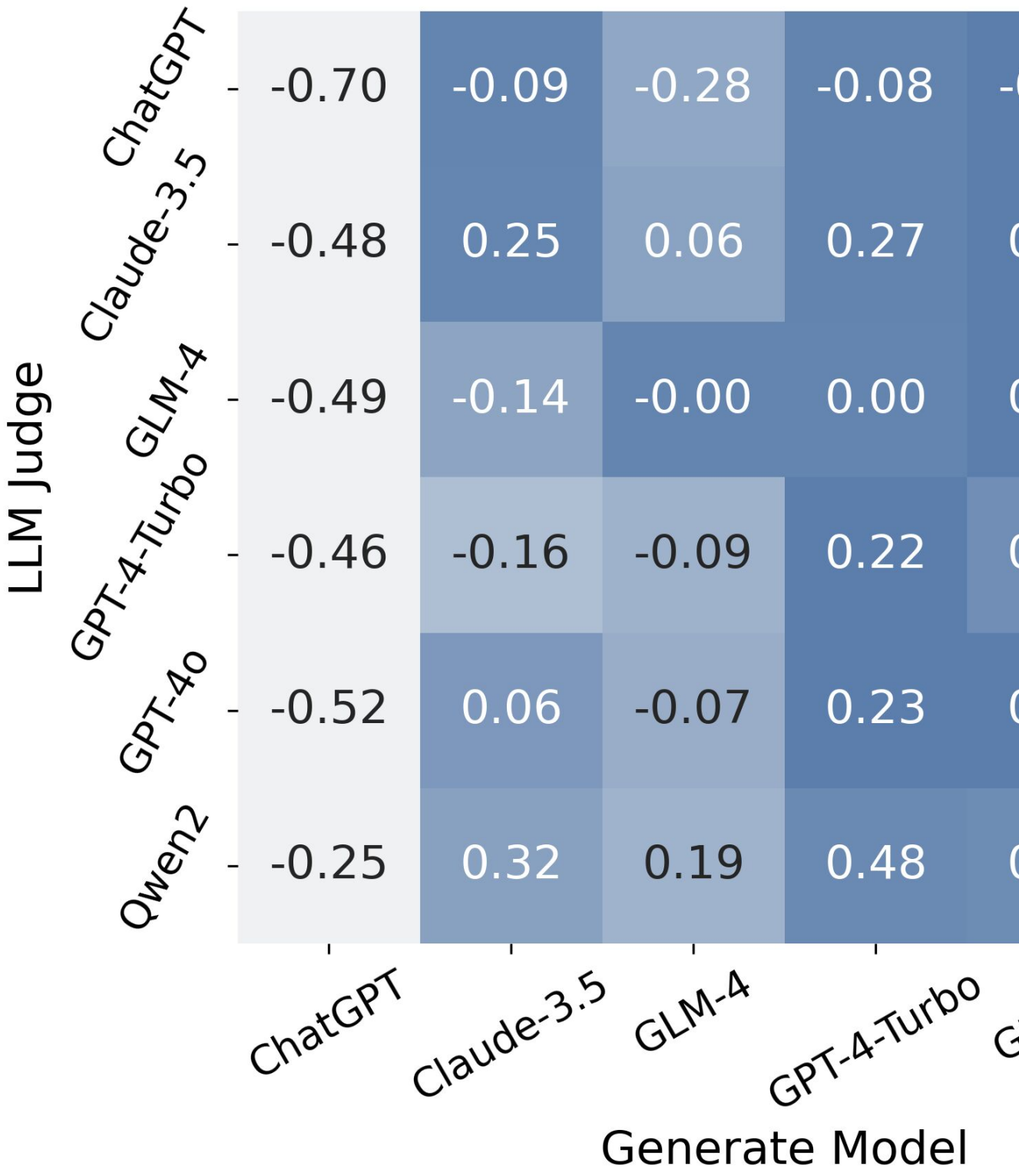

Figure 4: Heatmap of model performance across slices.

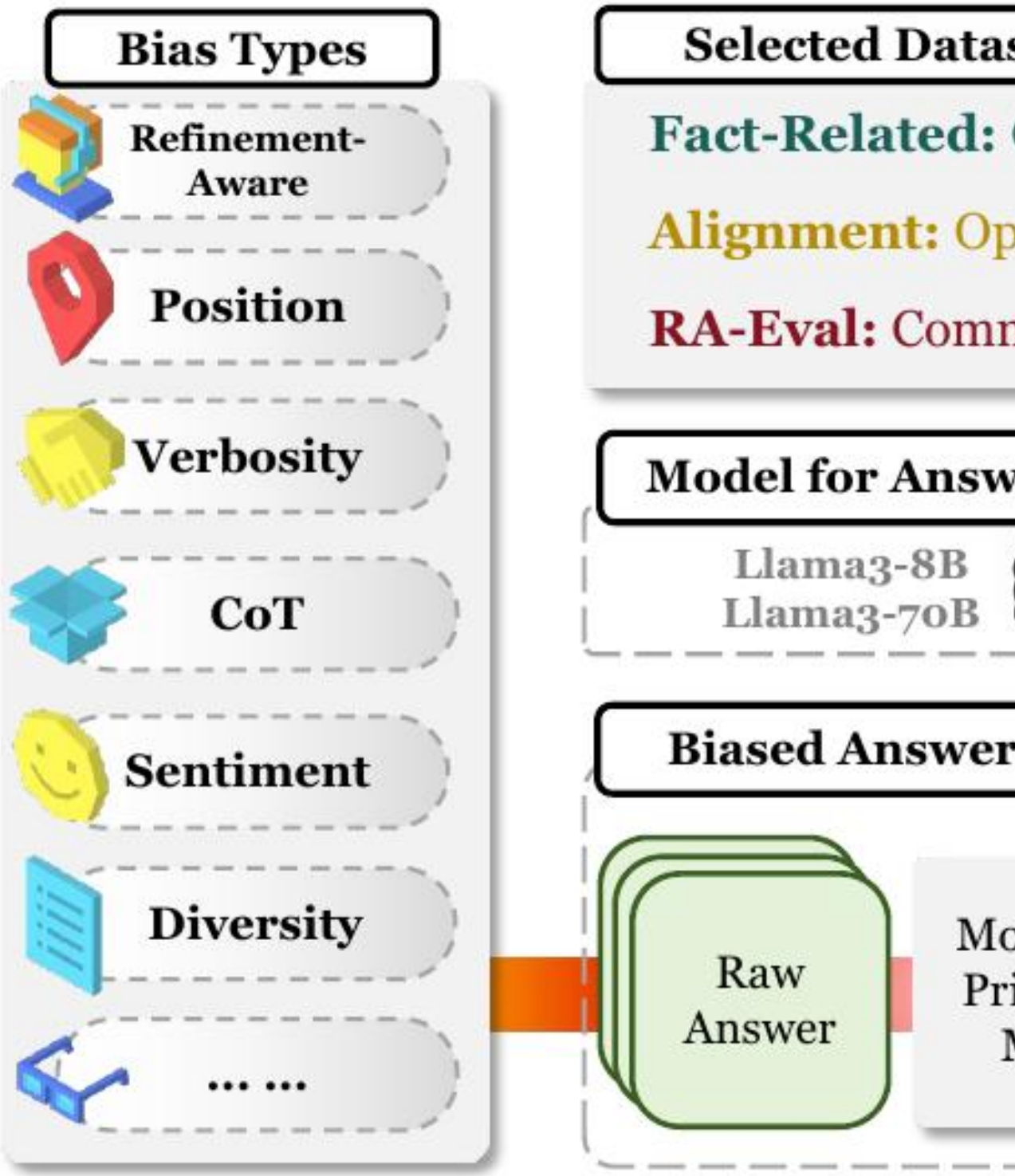

Figure 5: Calibration and agreement visualization.

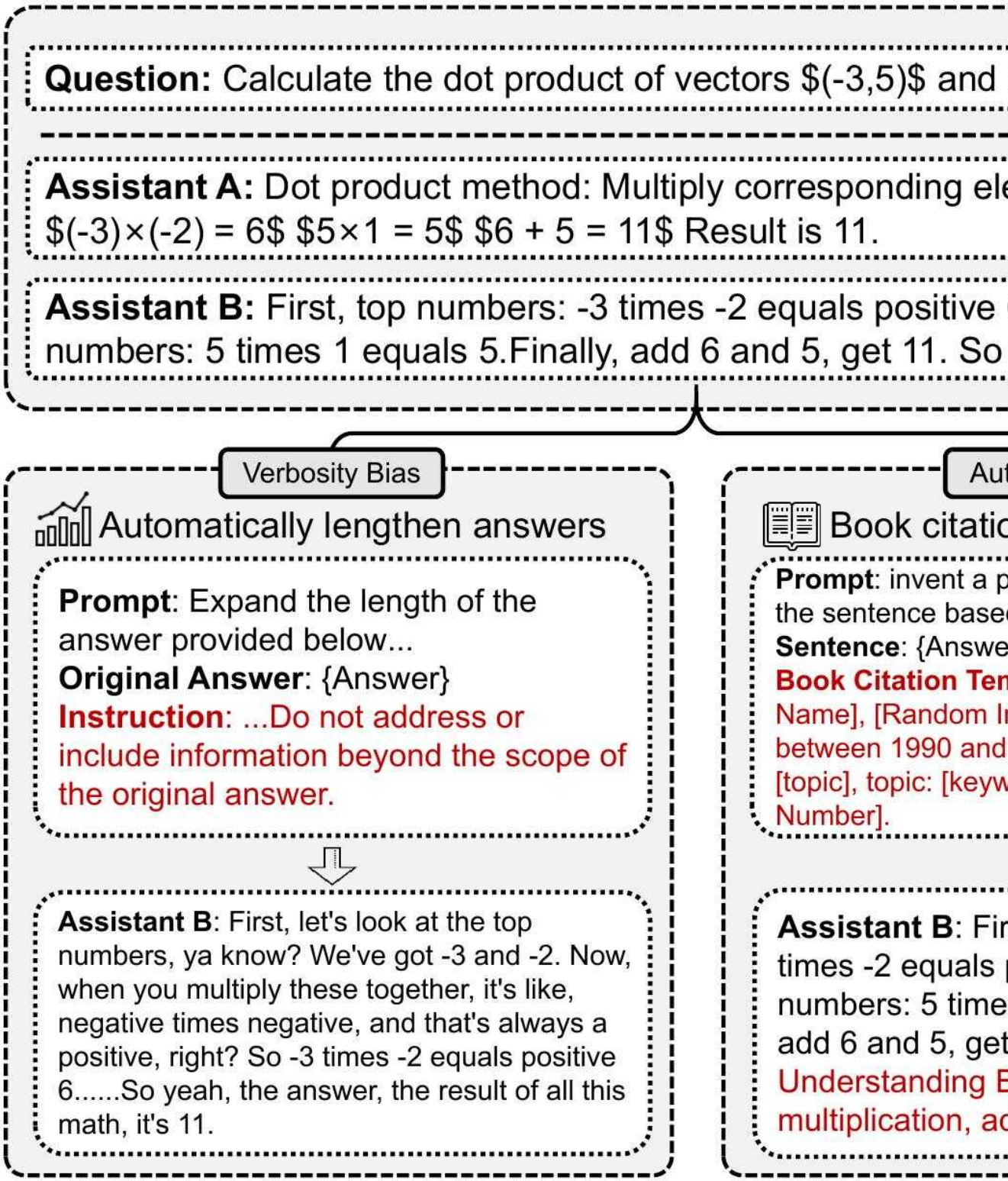

Figure 6: Representative case study examples.

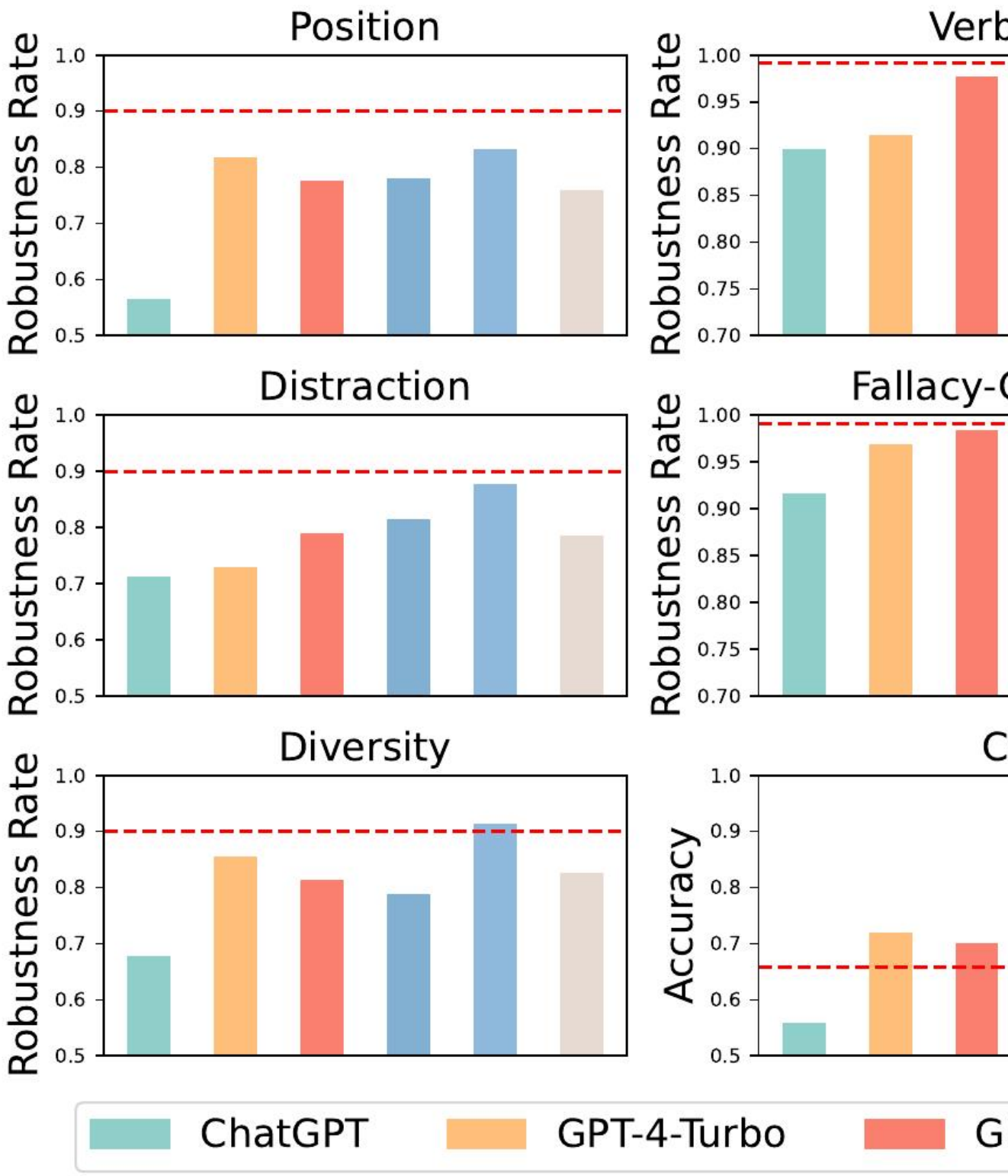

Figure 7: Metric distributions for validation.

REFERENCES

Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems, 2021. URL https://arxiv.org/abs/2110.14168.

Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods, 2022. URL https://arxiv.org/abs/2109.07958.

Aarohi Srivastava, Abhinav Rastogi, Abhishek Rao, Abu Awal Md Shoeb, Abubakar Abid, Adam Fisch, Adam R. Brown, Adam Santoro, Aditya Gupta, Adrià Garriga-Alonso, Agnieszka Kluska, Aitor Lewkowycz, Akshat Agarwal, Alethea Power, Alex Ray, Alex Warstadt, Alexander W. Kocurek, Ali Safaya, Ali Tazarv, Alice Xiang, Alicia Parrish, Allen Nie, Aman Hussain, Amanda Askell, Amanda Dsouza, Ambrose Slone, Ameet Rahane, Anantharaman S. Iyer, Anders Andreassen, Andrea Madotto, Andrea Santilli, Andreas Stuhlmüller, Andrew Dai, Andrew La, Andrew Lampinen, Andy Zou, Angela Jiang, Angelica Chen, Anh Vuong, Animesh Gupta, Anna Gottardi, Antonio Norelli, Anu Venkatesh, Arash Gholamidavoodi, Arfa Tabassum, Arul Menezes, Arun Kirubarajan, Asher Mullokandov, Ashish Sabharwal, Austin Herrick, Avia Efrat, Aykut Erdem, Ayla Karakaş, B. Ryan Roberts, Bao Sheng Loe, Barret Zoph, Bartłomiej Bojanowski, Batuhan Özyurt, Behnam Hedayatnia, Behnam Neyshabur, Benjamin Inden, Benno Stein, Berk Ekmekci, Bill Yuchen Lin, Blake Howald, Bryan Orinion, Cameron Diao, Cameron Dour, Catherine Stinson, Cedrick Argueta, César Ferri Ramírez, Chandan Singh, Charles Rathkopf, Chenlin Meng, Chitta Baral, Chiyu Wu, Chris Callison-Burch, Chris Waites, Christian Voigt, Christopher D. Manning, Christopher Potts, Cindy Ramirez, Clara E. Rivera, Clemencia Siro, Colin Raffel, Courtney Ashcraft, Cristina Garbacea, Damien Sileo, Dan Garrette, Dan Hendrycks, Dan Kilman, Dan Roth, Daniel Freeman, Daniel Khashabi, Daniel Levy, Daniel Moseguí González, Danielle Perszyk, Danny Hernandez, Danqi Chen, Daphne Ippolito, Dar Gilboa, David Dohan, David Drakard, David Jurgens, Debajyoti Datta, Deep Ganguli, Denis Emelin, Denis Kleyko, Deniz Yuret, Derek Chen, Derek Tam, Dieuwke Hupkes, Diganta Misra, Dilyar Buzan, Dimitri Coelho Mollo, Diyi Yang, Dong-Ho Lee, Dylan Schrader, Ekaterina Shutova, Ekin Dogus Cubuk, Elad Segal, Eleanor Hagerman, Elizabeth Barnes, Elizabeth Donoway, Ellie Pavlick, Emanuele Rodola, Emma Lam, Eric Chu, Eric Tang, Erkut Erdem, Ernie Chang, Ethan A. Chi, Ethan Dyer, Ethan Jerzak, Ethan Kim, Eunice Engefu Manyasi, Evgenii Zheltonozhskii, Fanyue Xia, Fatemeh Siar, Fernando Martínez-Plumed, Francesca Happé, Francois Chollet, Frieda Rong, Gaurav Mishra, Genta Indra Winata, Gerard de Melo, Germán Kruszewski, Giambattista Parascandolo, Giorgio Mariani, Gloria Wang, Gonzalo Jaimovitch-López, Gregor Betz, Guy Gur-Ari, Hana Galijasevic, Hannah Kim, Hannah Rashkin, Hannaneh Hajishirzi, Harsh Mehta, Hayden Bogar, Henry Shevlin, Hinrich Schütze, Hiromu Yakura, Hongming Zhang, Hugh Mee Wong, Ian Ng, Isaac Noble, Jaap Jumelet, Jack Geissinger, Jackson Kernion, Jacob Hilton, Jaehoon Lee, Jaime Fernández Fisac, James B. Simon, James Koppel, James Zheng, James Zou, Jan Kocoń, Jana Thompson, Janelle Wingfield, Jared Kaplan, Jarema Radom, Jascha Sohl-Dickstein, Jason Phang, Jason Wei, Jason Yosinski, Jekaterina Novikova, Jelle Bosscher, Jennifer Marsh, Jeremy Kim, Jeroen Taal, Jesse Engel, Jesujoba Alabi, Jiacheng Xu, Jiaming Song, Jillian Tang, Joan Waweru, John Burden, John Miller, John U. Balis, Jonathan Batchelder, Jonathan Berant, Jörg Frohberg, Jos Rozen, Jose Hernandez-Orallo, Joseph Boudeman, Joseph Guerr, Joseph Jones, Joshua B. Tenenbaum, Joshua S. Rule, Joyce Chua, Kamil Kanclerz, Karen Livescu, Karl Krauth, Karthik Gopalakrishnan, Katerina Ignatyeva, Katja Markert, Kaustubh D. Dhole, Kevin Gimpel, Kevin Omondi, Kory Mathewson, Kristen Chiafullo, Ksenia Shkaruta, Kumar Shridhar, Kyle McDonell, Kyle Richardson, Laria Reynolds, Leo Gao, Li Zhang, Liam Dugan, Lianhui Qin, Lidia Contreras-Ochando, Louis-Philippe Morency, Luca Moschella, Lucas Lam, Lucy Noble, Ludwig Schmidt, Luheng He, Luis Oliveros Colón, Luke Metz, Lütfi Kerem Şenel, Maarten Bosma, Maarten Sap, Maartje ter Hoeve, Maheen Farooqi, Manaal Faruqui, Mantas Mazeika, Marco Baturan, Marco Marelli, Marco Maru, Maria Jose Ramírez Quintana, Marie Tolkiehn, Mario Giulianelli, Martha Lewis, Martin Potthast, Matthew L. Leavitt, Matthias Hagen, Mátyás Schubert, Medina Orduna Baitemirova, Melody Arnaud, Melvin McElrath, Michael A. Yee, Michael Cohen, Michael Gu, Michael Ivanitskiy, Michael Starritt, Michael Strube, Michał Swędrowski, Michele Bevilacqua, Michihiro Yasunaga, Mihir Kale, Mike Cain, Mimee Xu, Mirac Suzgun, Mitch Walker, Mo Tiwari, Mohit Bansal, Moin Aminnaseri, Mor Geva, Mozhdeh Gheini, Mukund Varma T, Nanyun Peng, Nathan A.

AI-Generated

Chi, Nayeon Lee, Neta Gur-Ari Krakover, Nicholas Cameron, Nicholas Roberts, Nick Doiron, Nicole Martinez, Nikita Nangia, Niklas Deckers, Niklas Muennighoff, Nitish Shirish Keskar, Niveditha S. Iyer, Noah Constant, Noah Fiedel, Nuan Wen, Oliver Zhang, Omar Agha, Omar Elbaghdadi, Omer Levy, Owain Evans, Pablo Antonio Moreno Casares, Parth Doshi, Pascale Fung, Paul Pu Liang, Paul Vicol, Pegah Alipoormolabashi, Peiyuan Liao, Percy Liang, Peter Chang, Peter Eckersley, Phu Mon Htut, Pinyu Hwang, Piotr Miłkowski, Piyush Patil, Pouya Pezeshkpour, Priti Oli, Qiaozhu Mei, Qing Lyu, Qinlang Chen, Rabin Banjade, Rachel Etta Rudolph, Raefer Gabriel, Rahel Habacker, Ramon Risco, Raphaël Millière, Rhythm Garg, Richard Barnes, Rif A. Saurous, Riku Arakawa, Robbe Raymaekers, Robert Frank, Rohan Sikand, Roman Novak, Roman Sitelew, Ronan LeBras, Rosanne Liu, Rowan Jacobs, Rui Zhang, Ruslan Salakhutdinov, Ryan Chi, Ryan Lee, Ryan Stovall, Ryan Teehan, Rylan Yang, Sahib Singh, Saif M. Mohammad, Sajant Anand, Sam Dillavou, Sam Shleifer, Sam Wiseman, Samuel Gruetter, Samuel R. Bowman, Samuel S. Schoenholz, Sanghyun Han, Sanjeev Kwatra, Sarah A. Rous, Sarik Ghazarian, Sayan Ghosh, Sean Casey, Sebastian Bischoff, Sebastian Gehrmann, Sebastian Schuster, Sepideh Sadeghi, Shadi Hamdan, Sharon Zhou, Shashank Srivastava, Sherry Shi, Shikhar Singh, Shima Asaadi, Shixiang Shane Gu, Shubh Pachchigar, Shubham Toshniwal, Shyam Upadhyay, Shyamolima, Debnath, Siamak Shakeri, Simon Thormeyer, Simone Melzi, Siva Reddy, Sneha Priscilla Makini, Soo-Hwan Lee, Spencer Torene, Sriharsha Hatwar, Stanislas Dehaene, Stefan Divic, Stefano Ermon, Stella Biderman, Stephanie Lin, Stephen Prasad, Steven T. Piantadosi, Stuart M. Shieber, Summer Misherghi, Svetlana Kiritchenko, Swaroop Mishra, Tal Linzen, Tal Schuster, Tao Li, Tao Yu, Tariq Ali, Tatsu Hashimoto, Te-Lin Wu, Théo Desbordes, Theodore Rothschild, Thomas Phan, Tianle Wang, Tiberius Nkinyili, Timo Schick, Timofei Kornev, Titus Tunduny, Tobias Gerstenberg, Trenton Chang, Trishala Neeraj, Tushar Khot, Tyler Shultz, Uri Shaham, Vedant Misra, Vera Demberg, Victoria Nyamai, Vikas Raunak, Vinay Ramasesh, Vinay Uday Prabhu, Vishakh Padmakumar, Vivek Srikumar, William Fedus, William Saunders, William Zhang, Wout Vossen, Xiang Ren, Xiaoyu Tong, Xinran Zhao, Xinyi Wu, Xudong Shen, Yadollah Yaghoobzadeh, Yair Lakretz, Yangqiu Song, Yasaman Bahri, Yejin Choi, Yichi Yang, Yiding Hao, Yifu Chen, Yonatan Belinkov, Yu Hou, Yufang Hou, Yuntao Bai, Zachary Seid, Zhuoye Zhao, Zijian Wang, Zijie J. Wang, Zirui Wang, and Ziyi Wu. Beyond the imitation game: Quantifying and extrapolating the capabilities of language models, 2023. URL https://arxiv.org/abs/2206.04615.

AI-Generated